

Cross-session volatility persistence in XAUUSD

Gold Session Study II

Does Asia add information about London?

Direct OANDA bid/ask data | 2016-2023

Preregistered forecast comparison and independent-feed replication

Promising forecast improvement. Registered sample-size gate not met.

Adding Asian realised variance reduces validation QLIKE loss by 23.9% against the basic baseline and 20.0% against a stronger volatility-history baseline.

Only 160 development forecasts are available after the fixed training and data-quality rules, below the required 500. The conditional trading candidate was not run. The 2024+ holdout remains unopened.

XAUUSD Research project

Research and implementation assistance: Codex

4 October 2026 | gold-volatility-persistence-v1

Provider-delivered historical candles and hypothetical statistical forecasts. No live trades, profitable strategy, MT5 real-tick validation or COMEX replication are represented.

1. The result and the decision

A statistical lead survives; the acceptance gate does not.

Question	Actual result
Independent range-result replication	OANDA EXP001 coefficients are positive in development and validation; both HAC intervals exclude zero.
Asian RV -> London RV	Positive log-variance associations, including controls for prior London and daily variance.
Incremental validation forecasts	QLIKE improvement: basic 23.9%; stronger baseline 20.0%. MSE and MAE also improve.
Development adequacy	560 training observations through 2021; only 160 expanding-fit OOS forecasts, versus required 500.
Trading / final period	Not tested. Statistical acceptance is inconclusive under the fixed sample-size gate. Holdout remains locked.

The conclusion is not that the Asian feature contains no information. The observed forecast comparisons are favorable. The registered protocol requires more development forecasts before acceptance, and its requirement cannot be reduced after looking at validation.

The 23.9% loss reduction is not a trading return or profit percentage. Predicting volatility does not predict the first breakout side or guarantee post-entry travel.

2. Source provenance and acquisition

Source item	Observed record
Provider / instrument	OANDA v20 practice HTTPS API / XAU_USD
Resolution / price sides	Complete M1 candles, paired bid and ask; unsmoothed request
Requested history	1 January 2016 to 31 December 2023
Calendar response files	2,922; 439 contain no complete candles
Complete observations	2,789,384; incomplete observations 0
Raw response bytes	575,313,674, unchanged and SHA-256 verified
Quote audit	Invalid rows 0; flagged rows 4,893; genuine received both-side-flat bars retained 27,830

The API token was supplied through the local user environment. Only historical-candle GET requests were used. Source manifests contain public request URLs, UTC retrieval times, byte counts and hashes; they contain no token or account identifier.

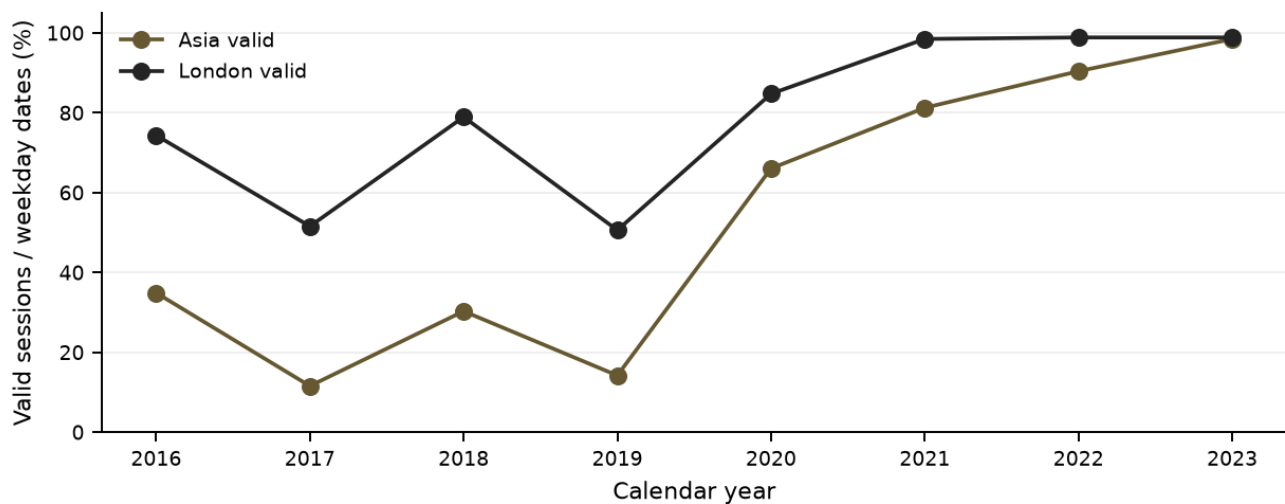
OANDA is the authenticated acquisition source for this study. It is a new broker feed relative to Study I's archive claiming Dukascopy origin. Direct Dukascopy authentication is still absent, so this is not two authenticated-provider replication.

Checksums establish the files used; provider HTTPS delivery establishes acquisition provenance. Neither guarantees that every historical quote is accurate or that historical venue pricing cannot be revised. Quote counts are provider price-update counts, not central gold trading volume.

3. Coverage is the binding limitation

Year	Weekdays	Asia valid	London valid	Daily valid	Forecast eligible
2016	261	91	194	183	87
2017	260	30	134	163	30
2018	261	79	206	203	78
2019	261	37	132	94	36
2020	262	173	222	146	141
2021	261	212	257	211	208
2022	260	235	257	214	228
2023	260	256	257	215	251

Received-bar eligibility changes across years



Source: OANDA 2016-2023; frozen M1/M5 completeness, boundary, gap and quote-quality rules. These are eligible data samples, not trading success rates.

A received minute is not automatically a valid M5 return block. Earlier years lose many sessions to missing M1 observations or insufficient complete M5 blocks. No missing price is interpolated. This creates a selected sample and limits generalisation, especially to quieter historical conditions.

Development supplies 560 eligible labelled training rows. The first 400 are used before the first development forecast on 11 March 2021; only the remaining 160 can be evaluated out of sample. Nominal development OOS dates were 2018-2021, but actual scored forecasts cover 11 March-17 December 2021.

4. Clocks, cleaning and causal labels

Window	Frozen local clock
Asia	09:00-15:00 Asia/Tokyo; UTC 00:00-06:00
London	08:00-12:00 Europe/London, historical DST
Daily control	Prior completed UTC weekday 00:00-24:00
New York	Secondary context only; no entry feature

All windows are half-open. Asia/London require 98% received M1 rows, both boundary minutes, maximum gap five minutes and no flagged quote; also 95% complete M5 blocks. The daily control uses 90% M1, 85% complete M5 and maximum gap 90 minutes.

Flag endpoint spread above \$5/oz and adjacent-minute midpoint jumps above 2%; reject invalid OHLC, nonfinite/nonpositive prices and nonpositive spreads. Genuine received flat candles are retained, with no interpolation or synthetic padding. Time gaps are measured as timedeltas; provider indices explicitly use nanoseconds.

Window	Invalid-date reasons
asia	insufficient_minutes: 717, insufficient_complete_m5: 230, flag_or_gap: 24, missing_boundary: 2
london	insufficient_minutes: 274, insufficient_complete_m5: 122, flag_or_gap: 27, missing_boundary: 4
daily	insufficient_minutes: 468, insufficient_complete_m5: 122, flag_or_gap: 67

Every forecast-origin eligible date stays in the forecast ledger even if its London target is subsequently missing. This run has zero missing targets among issued forecasts, but many input-ineligible dates. That does not establish unbiased coverage of all market regimes.

5. Reproducing Study I EXP001

Policy / split	N	EXP001 beta	95% HAC interval
flat exclusion / development	631	+0.273	[+0.178, +0.369]
flat exclusion / validation	464	+0.255	[+0.161, +0.349]
flat retained / development	786	+0.269	[+0.182, +0.355]
flat retained / validation	483	+0.308	[+0.204, +0.412]

EXP001 regresses log London percentage range normalised by its prior-20 median on log Asian dollar-range compression, with lagged Asian percentage-range baseline and prior absolute Asian return as controls. Each split resets rolling history, matching Study I.

The original prediction required a negative compression coefficient: quiet Asia followed by larger London ranges. OANDA reproduces a positive coefficient, supporting persistence rather than the registered expansion direction. This is range association, distinct from the variance forecast target in Study II.

The flat-exclusion diagnostic follows Study I's original received-bar deletion policy. The flat-retained version is the predeclared cleaning sensitivity. A separately labelled legacy millisecond-index run is saved; corrected and legacy gap screens give the same EXP001 estimates on these qualifying samples.

Shared-timestamp price differences with the archived feed are recorded separately. Different venues need not quote identical gold prices. Those differences do not authenticate the original archive or invalidate a reproduced statistical sign.

Replication supports the volatility research lead. It does not certify a compression-breakout strategy or replace a final unseen confirmation period.

6. Does the RV association replicate?

Period / model	N	Asian log-variance beta	95% HAC interval
development / simple	595	+0.597	[+0.494, +0.701]
development / enhanced	560	+0.365	[+0.282, +0.447]
development / HAR + Asia	560	+0.348	[+0.264, +0.432]
validation / simple	491	+0.782	[+0.682, +0.882]
validation / enhanced	479	+0.492	[+0.403, +0.581]
validation / HAR + Asia	479	+0.422	[+0.335, +0.508]

The simple association regresses log London realised variance on log current Asian realised variance. Both periods have a positive Asian coefficient and HAC interval above zero. The controlled models also retain a positive coefficient.

Realised variance is an unannualised sum of squared complete-M5 midpoint log returns. The first complete block uses its open-to-close return; subsequent close-to-close returns require adjacent blocks. Cross-gap returns are omitted. RV is its square root. No high/low-average midpoint extrema are invented.

Conditional validation regressions are association diagnostics only: their fitted coefficients never generate validation forecasts. Forecast coefficients and smearing stay fixed at their 2016-2021 estimates.

All intervals are conditional on the selected, observed sample and model assumptions. HAC allows specified short residual dependence; it does not establish causality, source independence, unbiased missingness or invariance across venues.

7. Registered forecast design

Model	Predictors, plus intercept
Basic baseline	log prior London variance; log prior daily variance
Basic + Asia	Basic baseline plus log current Asian variance
HAR-style baseline	Basic baseline plus logs of prior five- and twenty-session mean London variance
HAR + Asia	HAR-style baseline plus log current Asian variance

Target: log London realised variance. All four models use the same eligible dates and twenty-session history requirement. Prior controls use observations strictly before the forecast date, with at most four calendar days of staleness for prior London/daily variance.

Fit OLS with intercept. Convert the log prediction to variance by exponentiation times mean($\exp(\text{training residual})$); no validation residual enters this smearing adjustment. The fixed numeric forecast floor is $1e-12$. Realised targets are never clipped.

Development: expanding daily fits, minimum 400 prior eligible labelled observations. Validation: freeze coefficients and smearing at end-2021; only lagged features update causally. Were all gates met, final-period coefficients would be refitted once through 2023 and frozen before any holdout access.

The protocol predates forecast evaluation. Source-role and timestamp-correction amendments are documented separately and changed no model, loss threshold, percentile, stop, session hour or acceptance sample size.

8. Development out-of-sample results

Pair / metric	Baseline mean loss	Asia mean loss	Reduction
Basic / MSE	3.68e-11	2.73e-11	26.0%
Basic / MAE	4.63e-06	3.55e-06	23.2%
Basic / QLIKE	0.140	0.108	22.7%
HAR / MSE	3.03e-11	2.45e-11	19.1%
HAR / MAE	4.18e-06	3.3e-06	20.9%
HAR / QLIKE	0.121	0.096	20.6%

Scored period: 11 March-17 December 2021, 160 forecasts and 160 observed targets. Expanding estimation uses earlier completed sessions only. The nominal 2018-2021 test window produces no score before the 400-observation training minimum is reached.

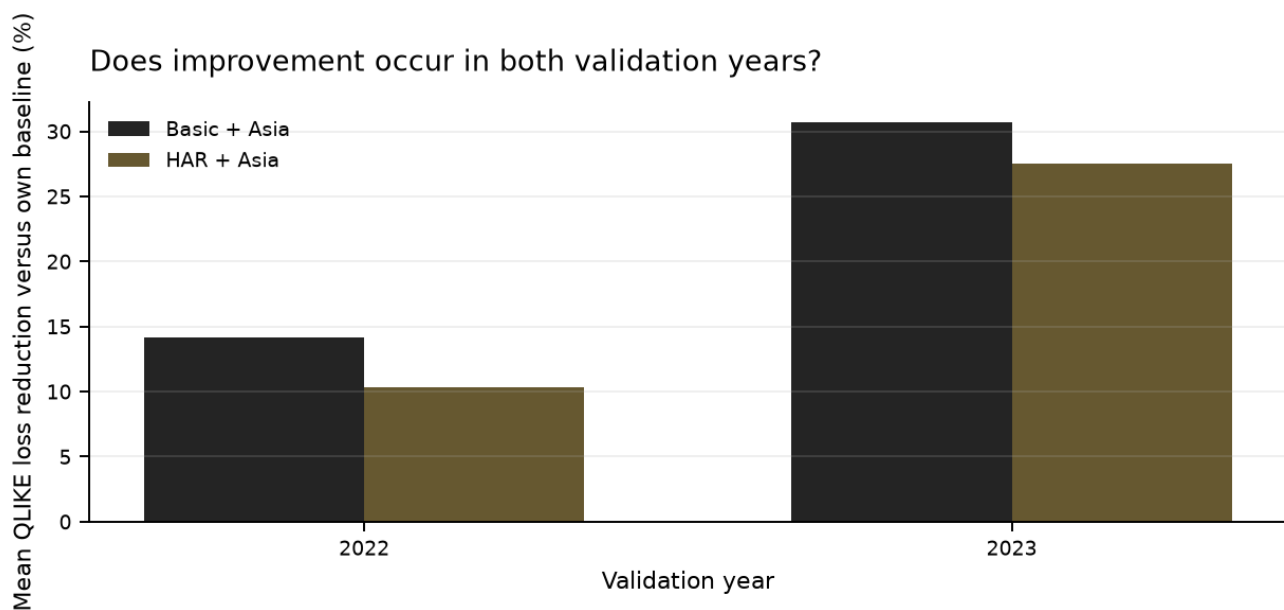
Both forecast pairs reduce all three mean losses in this available sample. This is evidence about 160 observed forecast dates, not the required 500 and not four fully evaluated development years.

MSE = $\text{mean}((\text{realised variance} - \text{forecast variance})^2)$; MAE = mean absolute variance error. QLIKE = $y/f - \ln(y/f) - 1$ for positive variance y and forecast f . Lower is better. Percentage reduction compares each enhanced model only with its corresponding baseline.

The training-window and completeness criteria remain unchanged. Lowering minimum training rows, relaxing received-bar rules or adding older history after seeing validation would be a new design requiring explicit disclosure and a clean evaluation plan.

9. Validation forecast losses

Pair / metric	Baseline mean loss	Asia mean loss	Reduction
Basic / MSE	6.62e-11	5.97e-11	9.8%
Basic / MAE	5.18e-06	4.34e-06	16.2%
Basic / QLIKE	0.162	0.123	23.9%
HAR / MSE	6.26e-11	5.82e-11	7.0%
HAR / MAE	4.87e-06	4.18e-06	14.3%
HAR / QLIKE	0.142	0.114	20.0%

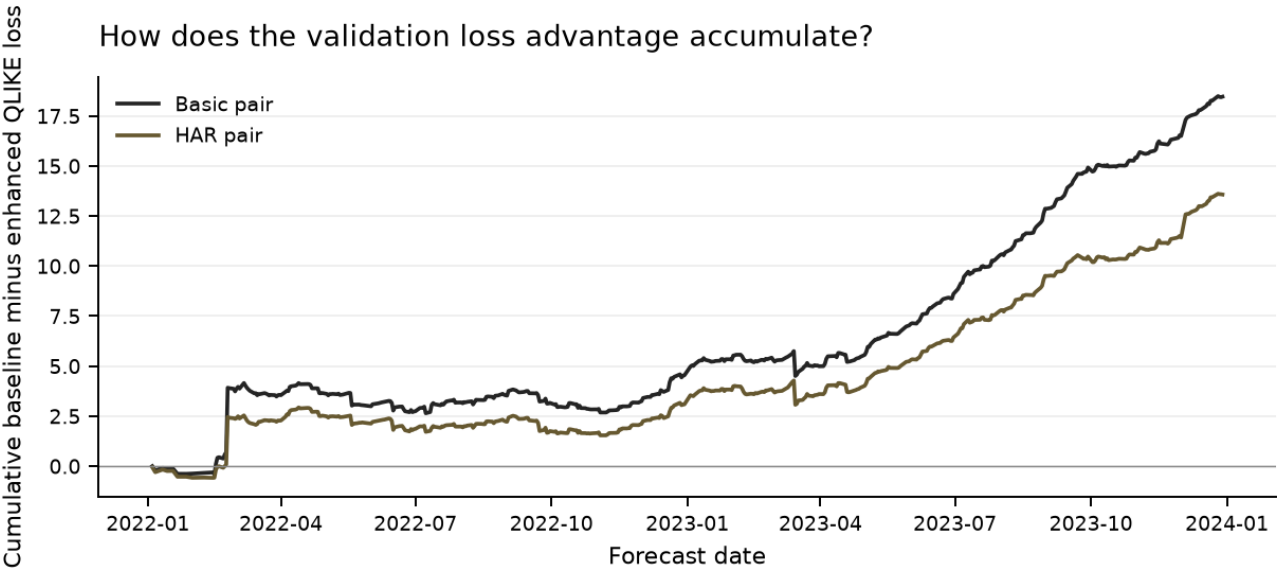


Source: OANDA matched forecasts, 2022-2023. Each model uses fixed 2016-2021 coefficients; lower forecast loss is not a financial return.

Validation covers 4 January 2022-29 December 2023, 479 issued and observed forecasts. Both model pairs improve QLIKE separately in 2022 and 2023. Validation MSE and MAE improve as well. No year or date was removed because its loss difference was unfavorable.

10. Uncertainty and model comparison

Validation pair	QLIKE difference	95% CI, 5-day block	95% CI, 10-day block
Basic	0.039	[0.022, 0.058]	[0.020, 0.060]
HAR	0.028	[0.014, 0.044]	[0.014, 0.045]



Source: 479 OANDA validation forecasts, 2022-2023. An evaluation-loss sum, not equity, PnL or an attainable trading payoff.

Paired circular blocks of five and ten forecast dates, 10,000 draws, seed 20261004. Positive lower intervals support lower mean enhanced-model loss in this observed sequence. The bootstrap preserves missing-label positions, although none occur among these issued forecasts.

The HAC loss-difference statistic is descriptive DM-style evidence. Nested models complicate standard DM inference. Secondary Clark-West tests operate on nested linear log forecasts, not smearing-adjusted QLIKE. Neither diagnostic overrides the minimum-sample gate or fully accounts for model-estimation uncertainty.

11. Acceptance gate and trading restraint

Acceptance condition	Basic pair	HAR pair
Development forecasts ≥ 500	FAIL	FAIL
Validation forecasts ≥ 200	PASS	PASS
Positive training Asian coefficient	PASS	PASS
Development QLIKE reduction $\geq 5\%$	PASS	PASS
Validation QLIKE reduction $\geq 3\%$	PASS	PASS
Both validation block CIs above zero	PASS	PASS
Both validation years improve QLIKE	PASS	PASS
MSE/MAE deterioration $\leq 5\%$	PASS	PASS

Decision: INCONCLUSIVE under the registered minimum-development-sample requirement.

Both model pairs fail the development count condition: 160 available versus 500 required. All other listed forecast conditions pass. A favorable validation result does not justify changing a failed predeclared sample-size criterion.

The single trading translation remains unrun: Asian RV strictly above the 80th percentile of its previous 60 valid sessions; first Asian bid-boundary break during London 08:00-10:00; opposite-edge executable-side stop; fixed London noon exit; no target and at most one trade.

Its registered scenario uses 0.25% risk, actual bid/ask spread, \$0.07/oz round-trip commission and \$0.05/oz adverse slippage per fill, plus doubled-cost stress. These are assumptions, not verified broker tariffs. Ordered tick replay and broker size/margin specifications are still absent.

No trade expectancy, profit factor, equity curve or Monte Carlo distribution is invented for Study II. There is no validated profitable strategy and no final-period access.

12. Study I timestamp erratum

A real implementation defect, preserved and corrected transparently.

Pandas 3 preserves millisecond resolution for epoch-millisecond conversions. Study I's gap check divided native index integers by a nanosecond minute constant, understating elapsed gaps. The intended five-minute rule was therefore not enforced correctly; the separate count, boundary and quote-flag rules still applied.

Period / diagnostic	Frozen trades	Corrected trades	Corrected PF / mean R
development / 0800 NY	732	732	0.921 / -0.039
development / 1200 NY	734	732	1.012 / +0.008
validation / 0800 NY	265	265	0.743 / -0.140
validation / 1200 NY	265	264	0.876 / -0.091

Re-running the existing archive with nanosecond indices leaves EXP001 unchanged: development beta +0.288493, N=1,508; validation beta +0.322187, N=495. The early-exit trade-R ledgers remain identical in both periods.

The corrected noon rule removes development NY-evening dates 19 August 2018 and 24 January 2019, and validation 17 October 2022. Its economic acceptance conclusion remains negative. Frozen noon ledgers and their Monte Carlo outputs describe the old screen and are not relabelled as corrected results.

The frozen report and archived bundle are unedited. Separate corrected ledgers and the implementation note provide the erratum. Study II now measures gaps using elapsed timedeltas and tests the same rejection across millisecond, microsecond and nanosecond indices.

13. Reproduction and evidence record

Validation comparison	DM-style HAC p, descriptive	Clark-West log-MSPE p
Basic	2.29e-05	1.36e-10
HAR	9.82e-05	6.85e-10

Core evidence files: SOURCE_INVENTORY.csv and SOURCE_SUMMARY.json; session_feature_and_coverage_ledger.csv; forecast_ledger.csv; development_coefficients.json; forecast_comparison.json; EXP001_direct_reproduction.json; ASSOCIATION_REPLICATION.json; corrected archive-gap feature/trade ledgers.

Registration: configs/study.json and preregistration/PROTOCOL.md. Dated implementation and source-role amendments document the timestamp repair and OANDA source choice before forecast evaluation. Original registration fingerprints remain intact.

Source/model code commit before evaluation: 8c42e39a0160c9745c27cb5bb866e7841a938ed9. Subsequent report-only commits may describe the finished deliverable; they do not replace the statistical registration.

Fourteen implementation checks pass: named-zone clocks, stale and strictly prior lags, integer time-delta parsing, paired complete quotes, missing-gap omission, future-label independence, retained flat candles, fixed validation coefficients, positive-target QLIKE and locked final dates. Synthetic test fixtures never enter empirical results.

Reproduce from this study folder: run the documented OANDA acquisition with a locally stored demo token; run forecast.py and replicate_exp001.py for provider oanda; run diagnostics.py for inventory, coverage and association records; run archive_gap_sensitivity.py using frozen Study I cached inputs; then rebuild this report. Public bundles exclude tokens, account identifiers and raw provider responses.

14. Conclusions and references

The research progressed from a rejected expansion hypothesis to a provider-supported volatility lead.

OANDA confirms the positive EXP001 range relationship and positive Asian-to-London realised-variance associations on qualifying data. Current Asian variance improves the registered validation forecasts against both explicit baselines.

Acceptance remains inconclusive because the development OOS sample is too small under the fixed rules. Source coverage is highly uneven across earlier years, and selected-session inference may not generalise to low-activity conditions. Neither this finding nor a volatility forecast alone establishes a profitable direction or execution rule.

The appropriate next work is to improve documented historical coverage or obtain another authenticated feed under a separately registered extension, without selecting criteria on viewed validation. Preserve 2024+ until all required gates and a final freeze justify opening it. No C++ optimisation or EA deployment is warranted by these results.

Primary methodology and source references

OANDA API access: developer.oanda.com/rest-live-v20/introduction

OANDA candle schema: OANDA v20 instrument specification

Pandas time resolution: [pandas 3.0 datetime-resolution documentation](https://pandas.pydata.org/pandas-docs/stable/timeseries.html)

Patton (2011): Volatility forecast comparison using imperfect volatility proxies

Clark and West (2007): Approximately normal tests for equal predictive accuracy in nested models

Motivation credit: TokyoCoil.pdf supplied the original regional-session case study; its USDJPY performance claims were not independently verified or used as evidence about gold. XAUUSD_Quant_Research_Master_Prompt.pdf supplied the research specification.

Exploratory research and statistical forecasts, not investment advice or a recommendation to trade. No live profitability, raw-tick execution proof, authenticated Dukascopy delivery or GC-futures validation is claimed.